

Equilibrium and non-equilibrium simulation studies of fluid alkanes in bulk and at interfaces

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State-of-the-art molecular dynamics (MD) methods are employed to study the properties of linear and branched fluid alkanes in bulk and at solid/fluid interfaces. The fluids are modelled with a transferable intermolecular potential, whose effectiveness is demonstrated through calculations of the shear viscosity using equilibrium and non-equilibrium techniques. Monolayer and multilayer films of linear and branched alkanes, on both metal and graphite surfaces, are studied to understand the effects of molecular architecture on the adsorption behaviour of these molecules. Preferential adsorption of a component in a mixture of linear and branched alkanes is also investigated. The parameters characterizing the hydrodynamic boundary conditions of confined liquid alkanes under shear flow are determined using linear response theory and non-equilibrium molecular dynamics (NEMD) simulations. The agreement between the present simulations on fluid alkanes and available experimental data is excellent.

1 Introduction

The study of complex fluids at interfaces has important practical implications for friction and lubrication.^{1–17} In this article, we focus on the application of modern computer simulation techniques to model lubricant systems, namely fluid alkanes. We first give details of the interaction potentials employed in the simulation, which are derived, in part, from phase diagrams. We then present results for bulk alkanes, both at equilibrium and under shear. Next, we study fluid alkanes on a surface, and explore the effects of molecular architecture on the properties of ultrathin films. In the final section, we investigate the behaviour of alkanes confined between two walls and characterize the hydrodynamic boundary parameters of the solid/liquid interface. Anticipating our results, we will see that MD is already able to reproduce properties of model lubricants quantitatively. Computer simulation is thus poised to begin a systematic study of more complex fluids and mixtures.

2 Alkane potentials

The alkane interaction potentials employed in all the simulations described here are of the united atom type, which treats methyl and methylene groups as pseudo-atoms or

beads bound together via harmonic springs:¹⁸

$$U_{\text{stretch}}(r_{ij}) = \frac{1}{2}k_r(r_{ij} - r_0)^2 \quad (1)$$

where r_{ij} is the distance between two bonded beads, k_r is the force constant to be fitted, and r_0 is the equilibrium bond distance. The potential used to describe the bond bending between three bonded atoms is also given by the simple harmonic form¹⁹

$$U_{\text{bend}}(\theta_i) = \frac{1}{2}k_\theta(\theta_i - \theta_0)^2 \quad (2)$$

where θ_i is the angle formed by the three bonded pseudo-atoms, k_θ is the force constant to be fit and θ_0 is the equilibrium angle. Torsions that include four bonded pseudo-atoms are described by a potential of the form,²⁰

$$U_{\text{torsion}}(\phi_i) = a_0 + a_1(1 + \cos \phi_i) + a_2[1 - \cos(2\phi_i)] + a_3[1 + \cos(3\phi_i)] \quad (3)$$

where ϕ_i is the torsional angle and a_i are the parameters to be fitted. The summary of all the parameters of the three intramolecular potentials are given in Table 1.

For interactions between non-bonded pseudo-atoms, we employed a pair-wise additive potential of the Lennard-Jones (LJ) form^{21,22}

$$U_{\text{LJ}}(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \quad (4)$$

Here, r_{ij} is the interatomic distance, σ_{ij} is the effective diameter of the methyl or methylene bead to be determined and ϵ_{ij} is the well depth. Lastly, some of the alkane molecules require a '1-5' non-bonded interaction of the form in eqn. (4), with different parameters.

The potential parameters for the bonded interactions were obtained from crystal structure data and from spectroscopic measurements. The parameters for non-bonded interactions were fit to experimental vapour-liquid coexistence curves. To do so, Gibbs ensemble Monte Carlo (GEMC)²³⁻²⁶ simulations were performed for linear and branched alkanes to obtain the coexistence curve and critical temperature.^{21,27} As an

Table 1 Table of bonded potential parameters^{21,27}

	$r_0/\text{\AA}$	$k_r/\text{K } \text{\AA}^{-2}$		
stretch	1.54	452 900		
bend	θ_0	k_θ/K		
$\widehat{\text{CH}}_2$	114°	62 500		
$\widehat{\text{CH}}$	112°	62 500		
$\widehat{\text{C}}$	112°	62 500		
torsions	a_0/K	a_1/K	a_2/K	a_3/K
$-\text{CH}_2-\text{CH}_2-$	0	355.03	-68.19	791.32
$-\text{CH}_2-\text{CH}-(\text{R})_2^a$	-251.06	428.73	-111.85	441.27
$-\text{CH}_2-\text{C}-(\text{R})_3$	0	0	0	461.29

^a (R)_i denotes that *i* alkyl groups are attached to the site.

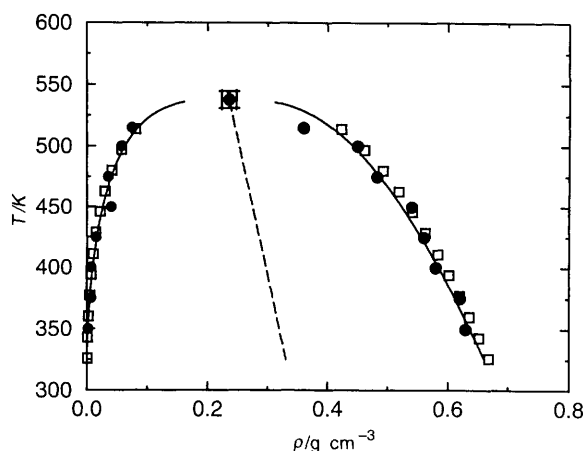


Fig. 1 Vapour-liquid coexistence curve of 3-ethylpentane. The CBMC/GEMC simulation data (this work) and experimental data (ref. 28) are depicted by circles and squares, respectively. The solid and dashed lines are fits to the scaling and rectilinear laws, respectively.

example, the coexistence curve for 3-ethylpentane is shown in Fig. 1. Such curves have also been computed for *n*-alkanes.^{21,27} Fig. 1 was used to tune the parameters for the CH group and the end segment for an ethyl side branch (see Table 2). The GEMC technique involves moves where molecules are inserted into either the liquid or vapour phase. Random insertion of a long-chain alkane into a dense phase is less probable and hence the configurational bias Monte Carlo (CBMC) technique was used to increase the sampling efficiency.^{29–32} Here, the inserted molecule is regrown bead by bead at a random position such that conformations of favourable energy are preferred. The resulting bias is offset by changing the acceptance rules in such a way as to satisfy the appropriate detailed balance condition.^{29–32}

The molecules considered in this article are *n*-decane, *n*-hexadecane, 6-pentylundecane, 7,8-dimethyltetradecane, 2,2,4,4,6,8,8-heptamethylnonane, *n*-triacontane and 2,6,10,15,19,23-hexamethyltetracosane (squalane). The force fields we employed are ‘transferable’ so that the same set of parameters has been used for all the fluids. The

Table 2 Table of non-bonded potential parameters^{21,27,a}

	ϵ/K	$\sigma/\text{\AA}$
CH ₃ long normal alkanes	114	3.93
CH ₃ ^o ethyl side-chain	97	3.93
CH ₃ methyl side-chain	78	3.93
CH ₂ all methylene groups	47	3.93
CH tertiary carbon	32	3.85

^a For HMN, non-bonded LJ parameters for the 1–5 interactions between beads in neighbouring branches employed $\sigma = 3.62 \text{ \AA}$ and $\epsilon = 40 \text{ K}$. Parameters for the 1–5 interactions in all other molecules, and all longer-ranged interactions in all of the molecules are given in the Table.

Table 3 Summary of results of the NPT MD calculation of densities of the branched and linear alkanes

liquid	<i>T</i> /K	cubic box length/Å	$\rho_{\text{sim}}^a/\text{g cm}^{-3}$	$\rho_{\text{exp}}^b/\text{g cm}^{-3}$
squalane	450	39.624	0.722	0.709
<i>n</i> -triacontane	450	39.686	0.719	not available
<i>n</i> -hexadecane	380	32.169	0.723	0.713

^a Each simulation was performed with 64 molecules. ^b The experimental density for squalane was extrapolated from ref. 35 whereas the value for *n*-hexadecane was taken from ref. 34.

parameters for both linear and branched alkanes, which were obtained by the fitting methods indicated above, are given in Tables 1 and 2.

The bulk fluid density calculated from a constant pressure MD simulation (NPT)³³ was used as a self-consistent check on the potentials. The system consisted of 64 *n*-hexadecane molecules at 380 K and 1 atm. For the C₃₀ isomers, squalane and *n*-triacontane, the simulations were performed with 64 molecules at 450 K and 1 atm. For both systems, long-range corrections to the pressure were included. In Table 3, the results are summarized and compared with experimental densities.^{34,35}

3 Shear viscosity

In the previous section, we developed potential parameters, which accurately reproduce both coexistence densities for liquid–vapour equilibrium and the corresponding critical point. In this section, we calculate the shear viscosity of the fluids by two different methods to test the effectiveness of the alkane potential.

3.1 Green–Kubo

The most common approach to the calculation of transport properties is to use Green–Kubo (GK) theory,^{37,38} which relates the property of interest to an appropriate time correlation function.³⁷ For example, the shear viscosity η can be written as

$$\eta(t) = \frac{V}{k_B T} \int_0^t \langle P_{xy}(0) P_{xy}(t^*) \rangle dt^* \quad (5)$$

where V is the volume, k_B is Boltzmann's constant, T is temperature, $\langle \cdots \rangle$ denotes an average over the canonical ensemble and P_{xy} is the xy component of the pressure tensor of the fluid.

The alkane *n*-decane was studied as a test case. The simulations were performed in the canonical ensemble on 64 molecules in a cubic box with edge length, 29.1 Å. A Nosé–Hoover Chain (NHC) thermostat of length five was coupled to all degrees of freedom to control the temperature.^{39–41} The NHC masses are given by the relation $Q_i = N_f k_B T \tau^2$,⁴¹ where N_f is the number of degrees of freedom and the time constant τ was set to 5 ps. Because long trajectories are needed to achieve converged results, a reversible version of the RESPA multiple time step algorithm was employed.^{42,43} The ‘fast’ timescale was determined by the intramolecular potentials and the ‘slow’ timescale by the LJ intermolecular potential. With this division of timescales, we were able to choose an outer (slow) time step of 3 fs and an inner (fast) time step of 1 fs. Each simulation was run for 300 000 outer time steps corresponding to 900 ps. Three indepen-

dent runs were performed at the state-point, $T = 480$ K and $\rho = 0.6136$ g cm⁻³ and the results for the viscosity were averaged.

In order to make connections with experiment, the pressure must be accurately computed and must include long-range corrections that are needed when the potential is truncated. The tail correction for a cut-off of 3.5σ for *n*-decane at the state-point in question is roughly 0.13 MPa. Results for the pressure and viscosity are given in Table 4.^{43,44}

Fig. 2 shows that the running integral $\eta(t)$ [eqn. (5)] is well converged after 120 ps, and the average of the three runs yields a viscosity that is in nearly quantitative agreement with experiment.^{43,44}

Simulations were also performed on 4-propylheptane, the star-shaped isomer of decane, and compared with experiment.⁴⁶ The conditions of the simulation were exactly the same as *n*-decane and the parameters were taken from Tables 1 and 2, so that direct comparisons of the viscosity, at the same density, can be made. The running integral, $\eta(t)$, is also shown in Fig. 2, and the summary of the results is given in Table 4.

Table 4 Summary of the simulation results for decane isomers

liquid	$\langle P \rangle^{a,b}/\text{MPa}$	$\langle \eta \rangle^a/10^{-6} \text{ Pa s}$	$n_{\text{exp}}^c/10^{-6} \text{ Pa s}$
<i>n</i> -decane	11.0 ± 1.0	190 ± 15	196.0
4-propylheptane	1.7 ± 0.8	160 ± 15	139.0

^a $\langle \rangle$ denotes a canonical ensemble average. ^b Pressures include tail corrections. ^c η_{exp} is the experimental viscosity, which for *n*-decane was taken from ref. 47. The experimental viscosity for 4-propylheptane was estimated from data taken from ref. 36 at 1 atm.

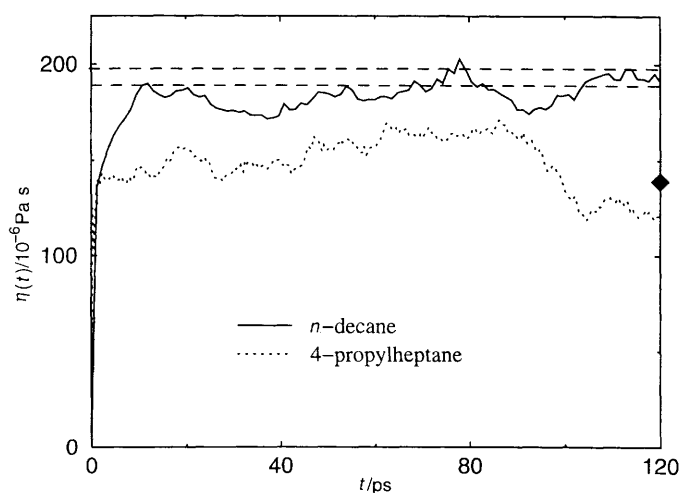


Fig. 2 Viscosity $\eta(t)$ for *n*-decane and 4-propylheptane averaged over three 900 ps NVT simulations at 480 K and 1 atm. The dashed line refer to the experimental value for *n*-decane taken from ref. 47 with bounds determined by the error bars in the calculated pressure.⁴⁵ The diamond is an Arrhenius extrapolation to 480 K and 1 atm of experimental data for 4-propylheptane taken from ref. 36.

The variation of viscosity with pressure at constant temperature is an important engineering parameter in the design of lubricants.⁴⁸ Usually one writes

$$\eta(P_2) = \eta(P_1)\exp[\alpha(P_2 - P_1)] \quad (6)$$

where, P_i is the pressure at the i th state point and α is the pressure-viscosity coefficient. The film thickness of a fluid under both load and shear can be determined theoretically with knowledge of α .⁴⁹ As a general rule, thicker films will act as better lubricants. Thus, an accurate determination of this property by simulation could be helpful in relating molecular architecture to lubricant performance.

With this in mind, we studied three additional state-points at respective densities of $\rho = 0.6382, 0.62511$ and 0.5937 g cm^{-3} . At each state point, a total of three independent simulations, each 900 ps in length, were carried out, starting from different well equilibrated initial configurations. From weighted linear fits to eqn. (6), the α coefficient for our model n -decane was determined to be $21 \pm 5 \text{ GPa}^{-1}$, compared to the experimental value of 14.3 GPa^{-1} (see Fig. 3).

Fig. 2 and 3 indicate that transport coefficients and thermodynamic data for alkanes can be calculated with reasonable accuracy from the parameter set in Tables 1 and 2. Given this finding, one can then proceed with reasonable confidence to compute the viscosity of materials that are more typical of real lubricants (say, isomers of C_{16} and C_{30}). Unfortunately, the GK method becomes much more difficult to converge for such large molecules because the relevant longest relaxation times increase. This, in turn, leads to prohibitively long simulation times if good statistics are to be achieved. A simple estimate shows that the end-to-end re-orientational relaxation time for n -decane at 480 K is roughly 10 ps. As a general rule, simulations must be run for two orders of magnitude longer than this time before good statistics can be obtained. For n -triacontane at 450 K, this relaxation time is *ca.* 200 ps. Thus, a GK simulation of this fluid would be prohibitively long with current computational resources. Accordingly, other methods must be utilized in order to compute the shear viscosity.

3.2 Non-equilibrium MD

The previous section was concerned with the calculation of the shear viscosity of alkanes using an 'indirect' method, *i.e.* via the equilibrium stress autocorrelation function.

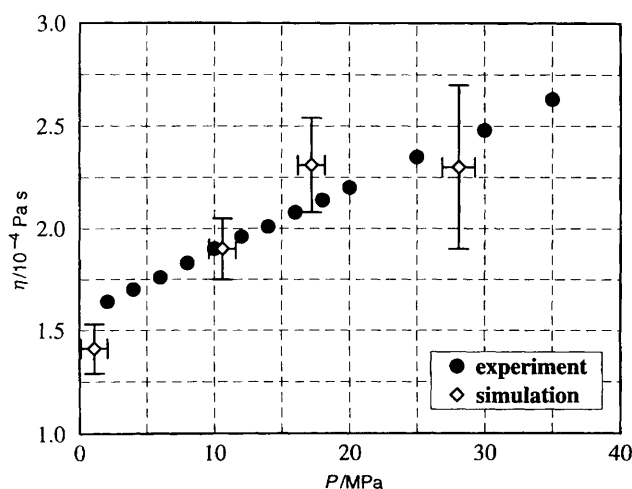


Fig. 3 Viscosity as a function of pressure for n -decane at 480 K. Experimental values are taken from ref. 47.

However, transport coefficients can be obtained directly through a set of equations, which within linear response, produce eqn. (5). The SLLOD equations for the case of planar Couette flow satisfy this requirement. The thermostatted SLLOD equations of motion are given by³⁷

$$\begin{aligned}
 \dot{\mathbf{r}}_i &= \frac{\mathbf{p}_i}{m_i} + \hat{\mathbf{i}}\gamma r_{yi} \\
 \dot{\mathbf{p}}_i &= \mathbf{F}_i - \hat{\mathbf{i}}\gamma p_{yi} - \mathbf{p}_i \frac{p_{\xi 1}}{Q_1} \\
 \dot{\xi}_i &= \frac{p_{\xi i}}{Q_i} \\
 \dot{p}_{\xi i} &= \sum_{i=1}^N \frac{p_i^2}{m_i} - N_f k_B T - p_{\xi 1} \frac{p_{\xi 2}}{Q_2} \\
 \dot{p}_{\xi j} &= \left(\frac{p_{\xi j-1}^2}{Q_{j-1}} - k_B T \right) - p_{\xi j} \frac{p_{\xi j+1}}{Q_{j+1}} \\
 \dot{p}_{\xi M} &= \left(\frac{p_{\xi M-1}^2}{Q_{M-1}} - k_B T \right)
 \end{aligned} \tag{7}$$

Here, $\mathbf{r}$ and $\mathbf{p}$ are the position and momentum, respectively, j denotes the index for a NHC chain of length M , N is the total number of particles, N_f is the number of degrees of freedom, $\hat{\mathbf{i}}$ denotes a unit vector in the x direction, γ is the shear rate and ξ and p_ξ are the NHC position and momentum. Linear response theory, with P_{xy} being the dependent variable and γ the independent variable, yields³⁷

$$\langle P_{xy} \rangle_{\text{NE}} = - \frac{V}{k_B T} \int_0^\infty \langle P_{xy}(t) P_{xy}(0) \rangle \gamma \, dt \tag{8}$$

Here, the subscript NE indicates the steady state non-equilibrium ensemble produced by eqn. (7). By invoking eqn. (5), one obtains

$$\eta = - \frac{\langle P_{xy} \rangle_{\text{NE}}}{\gamma} \tag{9}$$

Thus, one performs a simulation using eqn. (7) with a particular γ and calculates $\langle P_{xy} \rangle_{\text{NE}}$ in the steady state. The GK shear viscosity is given by eqn. (9) in the limit $\gamma \rightarrow 0$.

The important advantage of this method is that one does not have to converge the integral of an autocorrelation function, but rather converge an average which, in principle, should require less CPU time. However, extrapolating to zero shear is problematic. To date, there has not been a theory that has achieved this in a satisfactory fashion for a general molecular fluid.⁵⁰ Thus, one is left with the problem of finding a γ that is in the Newtonian regime of the fluid, where the shear viscosity becomes independent of γ . This ‘plateau’ region is where one can safely extrapolate to the GK value.^{50,51}

We chose to perform both constant pressure (NPT-SLLOD) and constant volume (NVT-SLLOD) NEMD simulations. The starting states for both types of runs were well equilibrated configurations of 64 *n*-decane molecules at 480 K at $\rho = 0.6136 \text{ g cm}^{-3}$. The details of the algorithms are discussed in ref. 44, 51, 52. Fig. 4 shows that the NVT-SLLOD and NPT-SLLOD converge at low shear rates γ . It is also interesting to note the onset of a plateau around $\gamma = 0.1 \text{ ps}^{-1}$, from which we are able to extrapolate to the GK value, within the combined error bars.⁵¹ Note that in the shear-thinning

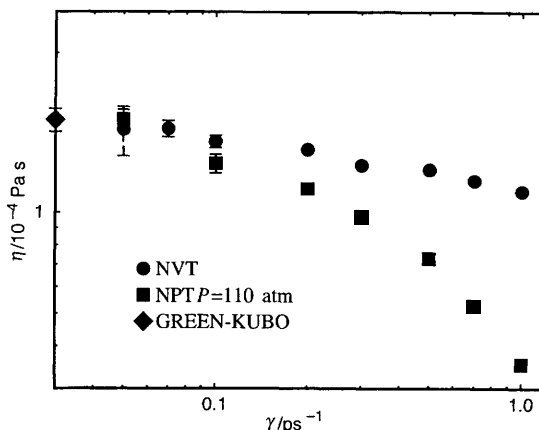


Fig. 4 Dependence of viscosity $\eta(\gamma)$ on the shear rate γ for *n*-decane at 480 K and $P = 110$ atm using NPT-SLLOD and NVT-SLLOD. The Green-Kubo result is placed on the y -axis as a guide for the eye.

regime of Fig. 4, the NVT and NPT-SLLOD simulations differ. At NVT conditions, the system is not able to relieve the pressure increase by adjusting the density.⁵¹

To investigate the structure of the fluid under shear, we examine the distributions for components of the molecular end-to-end vector. Fig. 5 shows this distribution at equilibrium. There is no preferred orientation of the end-to-end vector and the fluid is indeed isotropic. Fig. 5 also shows the distribution of end-to-end vectors for NVT- and NPT-

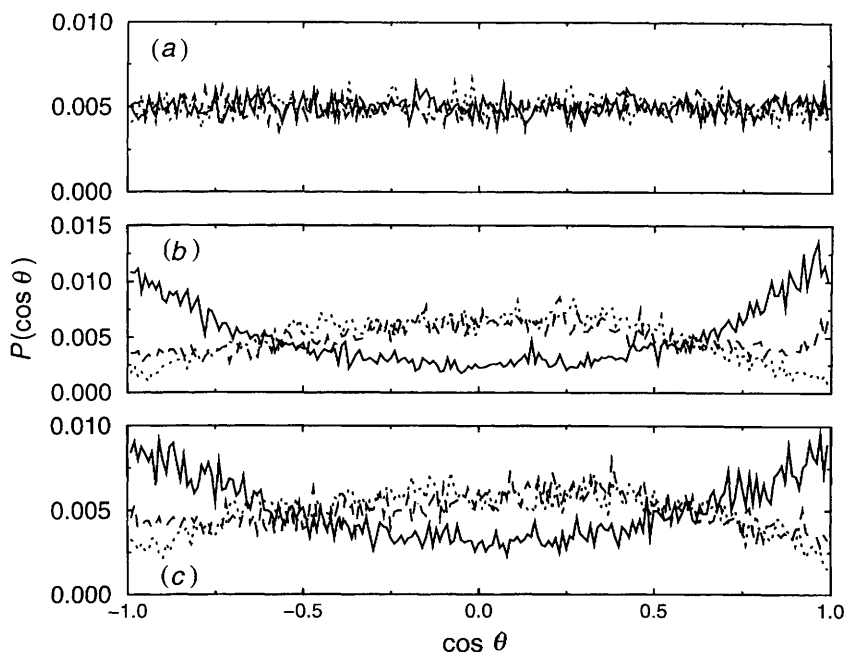


Fig. 5 Distribution of molecular end-to-end projection vectors for an NVT simulation of *n*-decane at 480 K, (a), NVT-SLLOD simulation of *n*-decane at 480 K and $\gamma = 0.5$ ps⁻¹, (b), and an NPT-SLLOD simulation of *n*-decane at 480 K and $\gamma = 0.5$ ps⁻¹, (c). Solid lines denote projections on the x -axis, dotted lines are projections on the y axis and dashed lines are projections on the z -axis.

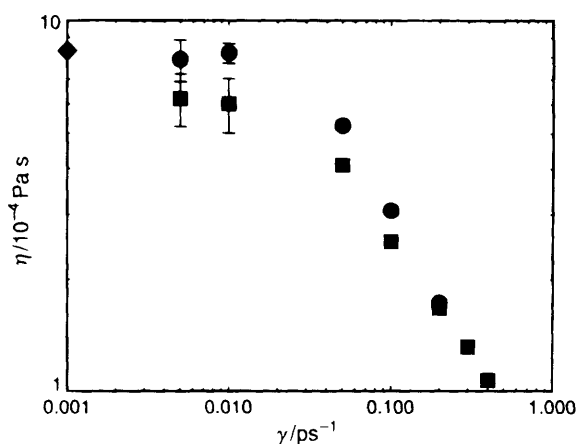


Fig. 6 Viscosity of *n*-hexadecane (■) and 2,2,4,4,6,8,8-heptamethylnonane (●) under shear using the NPT-SLLOD algorithm at 380 K and $P = 1$ atm. The ♦ on the y -axis is the experimental value for *n*-hexadecane ($\gamma = 0$) at $\rho = 0.72 \text{ g cm}^{-3}$. As expected, the viscosity of branched isomers is higher at the same state point.

SLLOD, at $\gamma = 0.5 \text{ ps}^{-1}$. In the former case, there is more shear alignment in the direction of flow (x direction). This difference arises because the only mechanism that is available to relieve pressure in the NVT simulation is alignment with the flow, whereas NPT-SLLOD allows the density to change as well.

Since reasonable confidence has been built in our ability to simulate *n*-decane, it is worthwhile to study the effects of branching. To start, we would like to work at densities and temperatures that are accessible to both simulation and experiment. We chose to study the isomers of C_{16} , namely *n*-hexadecane and 2,2,4,4,6,8,8-heptamethylnonane. Even though there is no experimental data available for the latter, interesting comparisons can be made with *n*-hexadecane because the bulk density is similar to *n*-hexadecane at $P = 1$ atm. Both isomers were simulated at 380 K and $P = 1$ atm using the NPT-

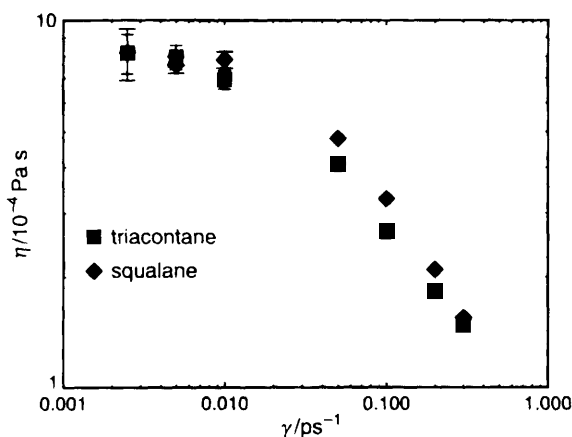


Fig. 7 Viscosities of *n*-triacontane and squalane as a function of shear rate

SLLOD algorithm, which is described in ref. 51. Experimental results for the $\gamma \rightarrow 0$ limit of the shear viscosity are given in ref. 34 and a comparison is made in Fig. 6. It is clear that the onset of a plateau is approached for $\gamma \leq 0.1 \text{ ps}^{-1}$. Simulations for $\gamma \leq 0.01 \text{ ps}^{-1}$ become prohibitively long for C_{16} . Results for $\eta(\gamma)$ for 2,2,4,4,6,8,8-heptamethylnonane which are also shown in Fig. 6, indicate that a higher shear viscosity than that for *n*-hexadecane, is obtained in the limit that $\gamma \rightarrow 0$.

Simulations were also performed for *n*-triacontane and squalane at 450 K and $P = 1$ atm. Again, because no experimental data is available we chose to plot the data against the C_{16} results at 380 K and $P = 1$ atm. This is shown in Fig. 7. At higher shear rates ($\gamma \geq 0.1 \text{ ps}^{-1}$), all the viscosities seem to be in qualitative agreement. The simulations show that for C_{30} isomers, the $\gamma \rightarrow 0$ shear viscosity is in qualitative agreement with *n*-hexadecane at 380 K. Unfortunately, it is difficult to see a statistical difference between the data for triacontane and squalane.

4 Liquid/solid interface

The behaviour of molecular fluids under confinement is different than in bulk.^{1–17} The force between two surfaces mediated through a fluid exhibits well defined oscillations, which are associated with the layering of fluid molecules.³ Typically, these density oscillations persist until *ca.* 20 Å above the surface, although the details depend on the specific surface–fluid interaction potential. Surface forces have been experimentally studied using the surface force apparatus (SFA).⁵³ Correspondingly, density oscillations in confined fluids have been studied through MD and Monte Carlo simulation methods.^{4,9,10,12,54–58} Both linear and branched molecules form layers against a surface, though to varying degrees.^{59,60} Confined fluid films, with thickness of the order of a few molecular diameters, exhibit both enhanced viscosity and relaxation times and show yield stress properties typical of a solid.^{6,7,14,52,61–63}

4.1 Equilibrium studies

Density oscillations in a fluid near a surface can be studied by computer simulations. Bead density profiles of liquid *n*-hexadecane films adsorbed on a Au(111)¹⁷ surface are shown in Fig. 8. A gradual development of the density oscillations is observed as the thickness of the film is increased. The density profile of the thickest film shows four clear oscillations followed by a uniform density representing the bulk. The average density in the bulk region of the film (between 25 and 35 Å away from the surface) matches well with the experimental equilibrium bulk density of *n*-hexadecane at 350 K (0.73 g cm^{-3}).³⁴ The alkane potential model,^{21,22,27} is thus found to reproduce bulk properties fairly successfully. The features of the first peak in the density profile are unaltered by film thickness, implying that the average density in the first layer remains the same irrespective of the thickness of the film. The average density of the first layer is around 40% higher than the density of bulk hexadecane liquid at 350 K.

Subtle features in the structure of adsorbed molecules can be observed if the adsorption is commensurate with the underlying lattice of the surface. This is true for alkanes adsorbed on graphite. The equilibrium distance between next-nearest-neighbour beads in an alkane chain is around 2.54 Å, which is nearly equal to the lattice constant of graphite (2.46 Å). Thus, monolayers of alkane and alkanol molecules have been observed to form lamellar structures on graphite.^{64,65} Scanning tunnelling microscope (STM) experiments^{66–68} have revealed that the alkane molecules are preferentially adsorbed on the surface with their backbone axis parallel to it. At sufficiently low temperatures this facilitates the formation of well ordered arrays of chains on the surface. The temperature

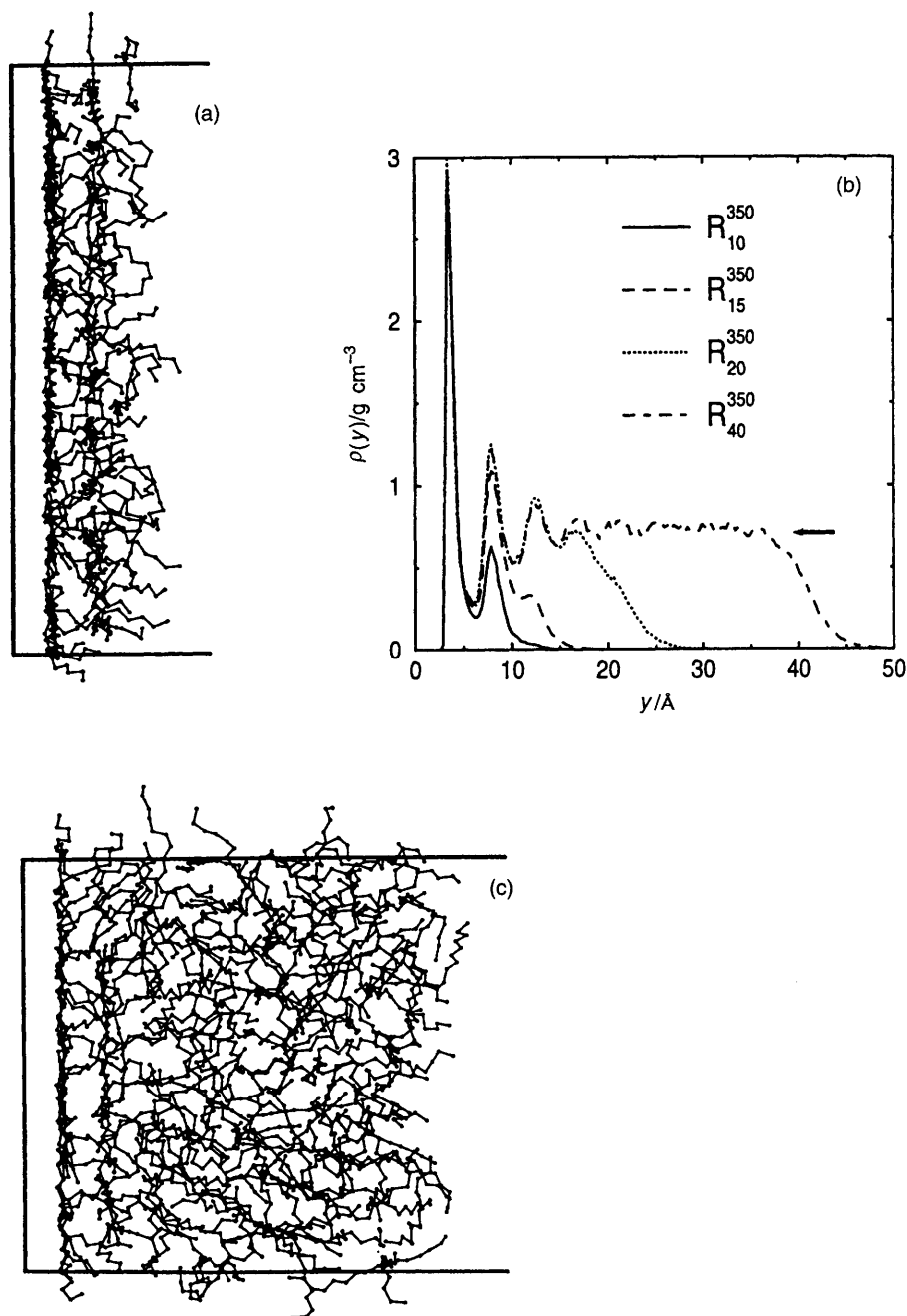


Fig. 8 Bead density profiles normal to the surface (y -coordinate) for n -hexadecane films on Au(111) at 350 K. Films of different thicknesses were studied through the CBMC method.^{29–32} The density profile (b) is sandwiched between single configurations of the final configuration of a 15 Å (a) and 40 Å (c) film. The bulk density of n -hexadecane at 350 K is marked by an arrow. Runs of different films are denoted by R_t^T , where T is the temperature in Kelvin and t is the thickness of the film.

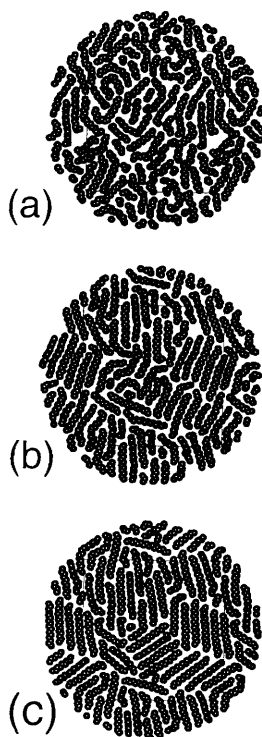


Fig. 9 Snapshots of the top view of *n*-hexadecane molecules in the first layer on a graphite surface at (a) 350, (b) 300, (c) 250 along with their periodic images for a 15 Å thick film. Bead sizes correspond to their van der Waals radii. The edge length of the square box is 61.2 Å. The graphite–alkane interactions were modelled through the Steele potential.^{70,71} Films were studied by the CBMC.¹⁷ The degree of in-plane ordering increases as a function of decreasing temperature. Well ordered lamellae are seen at 250 K. The zig-zag nature of the conformations of molecules indicates that the backbone plane is parallel to the graphite surface in these films.

dependence of the structure of domains of such alkanes (roughening of domains) has also been studied.⁶⁷ Single configurations of the first layer from our simulation⁶⁹ of a *n*-hexadecane film on graphite at three temperatures are shown in Fig. 9. The structure of the first layer at 350 K is similar to the one on Au(111). Organized lamellae begin to form at 300 K and are more evident at 250 K.

The density profiles of the films are shown in Fig. 10. The height of the first peak increases with decreasing temperature indicating denser packing of molecules. This is also supported by the small decrease in the separation between the peaks in the density profile. Close packing of molecules in the first layer of such films, both on Au(111) and on graphite, is achieved by an increase in their end-to-end length.¹⁷ Thus, the molecules in the first layer have fewer *gauche* defects than those in the bulk. For example, at 350 K, *ca.* 80% of the dihedral angles of molecules in the first layer are *trans*, while it is *ca.* 68% for molecules in the bulk region of the film.¹⁷

The orientation of the backbone zig-zag plane of the alkane molecules on graphite has been somewhat unclear;^{66–68,72} STM results of long chain alkanes (ranging from C₂₁H₄₄ to C₃₂H₆₆) have been interpreted in different ways. The zig-zag plane has been conjectured to be either parallel or perpendicular to the substrate. The parallel orientation provides a higher surface density of the chains than the perpendicular orientation.

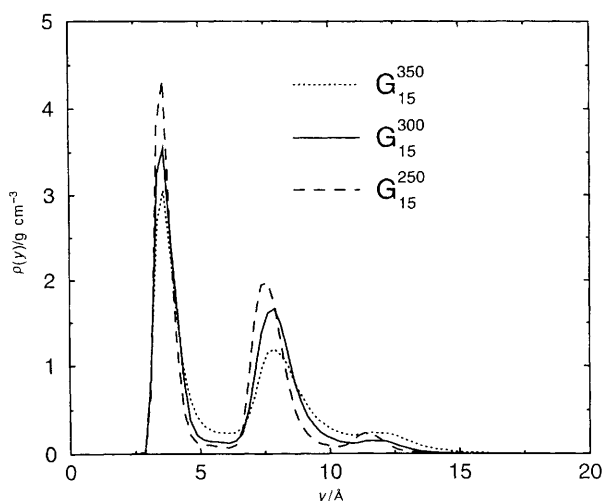


Fig. 10 Bead density profiles normal to the graphite surface (y -coordinate) for films shown in Fig. 9. Curves are labelled by G_t^T , where T is the temperature in K and t is the thickness of the film.

The important difference between the two orientations is that the intermolecular spacing in the parallel orientation is somewhat larger than the lattice spacing of graphite, so that long-range commensurability with the lattice is lost; whereas when alternate molecules have a perpendicular orientation, they regain the long range commensurability with the surface, even though locally these molecules may not be favourably aligned with the surface.⁷² Recent neutron scattering experiments⁷² have indicated the importance of solvent effects on the in-plane structure of these films. The neutron scattering data of a monolayer of $C_{32}H_{66}$ in a C_7H_{16} solvent has been found to fit a model of chains with alternating parallel and perpendicular orientations. Whereas, a configuration of alkane molecules with their zig-zag planes parallel to the substrate fits the neutron profile data of a vapour deposited monolayer of the long chain alkane.⁷² In our simulations of hexadecane films, we find that the molecules tend to adsorb with the zig-zag planes parallel to the graphite surface. This is consistent with the experimental results obtained for the vapour deposited monolayer.⁷² That the backbone zig-zag plane of molecules in the film is oriented parallel to the surface is also seen from the single configurations of the molecules in the first layer as shown in Fig. 9.

The structure of alkane molecules near a solid surface depends on its internal architecture. It is known from SFA experiments⁵⁹ and MD simulations^{15,73} that films of branched molecules also exhibit oscillations in the surface force just as those of linear molecules. Subtle differences in layering do exist. Simulations of 2-methylheptane^{15,73} indicated density oscillations, while earlier SFA experiments on 2-methyloctadecane and 2-methylundecane had no oscillations in the surface force profile.⁷⁴ More recent SFA experiments have confirmed the MD results.⁵⁹ We have carried out simulations of films of isomers of hexadecane on an Au(111) surface.⁶⁰ The density profile of such films, studied under identical conditions, are compared in Fig. 11. In addition to the linear alkane, three branched isomers have been studied: 6-pentylundecane (PUD) a star-shaped molecule, with equal lengths in each branch; 7,8-dimethyltetradecane (DMT) a long-chain molecule, with two single pendant methyl groups at the central region of the backbone and 2,2,4,4,6,8,8-heptamethylnonane (HMN) a much more rigid molecule, with branches on all the beads of the backbone. Branching decreases the extent of close packing, and thus ordering, on the surface. This is seen as a decrease in the height of the first peak in the density profile with increasing number of branches.

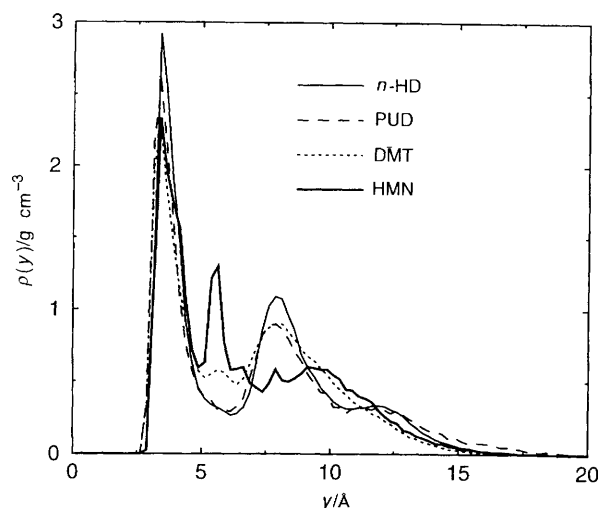


Fig. 11 Bead density profiles of thin films of *n*-hexadecane (*n*-HD), 6-pentylundecane (PUD), 7,8-dimethyltetradecane (DMT) and 2,2,4,4,6,8,8-heptamethylnonane (HMN) at 350 K. Each film contained 72 molecules on Au(111). Details of the simulation can be found in ref. 60.

From an analysis of the contribution of the bead types to the total bead density profile, it is possible to understand the geometry of the adsorbed molecule on the surface. In the case of the PUD film, we find that the central bead is shifted above the first layer to accommodate all three arms in the first layer. The rigid HMN molecule packs irregularly on the surface and has only around six of its beads in the first layer. The height of the first peak in the density profile is thus decreased. The contribution from the branches of the HMN molecule leads to a new peak between the first and the second layer. This peak is also observed for DMT. The presence of extensive branches in HMN effectively increases the separation between layers. Thus, the peaks in the density profile for HMN (and to some extent for DMT) are separated by *ca.* 6 Å, while for the linear isomer and for PUD it is *ca.* 5 Å. The minimum between the first and second peaks in the density profile decreases as a function of branching. Branched molecules belonging to the first layer interdigitate more than linear molecules, increasing the density of beads between layers. Interdigitation is believed to influence rheological behaviour of confined alkane films, thereby having a direct bearing on lubrication.⁶⁰ Fig. 12 shows single configurations of molecules belonging to the first layer for the thicker film. While *n*-hexadecane and PUD molecules interdigitate less, a significant proportion of HMN molecules extend their branches into the second layer.

Surfaces can also separate components from a multicomponent mixture by selective adsorption of a particular species.⁷⁵ Similarly, the vapour/liquid interface can also enrich a particular species. Segregation of species leads to decreased entropy, but could lead to a decrease in the free energy of the system through an increased interaction energy with the surface. Most commonly used lubricants are usually multicomponent mixtures of long chain alkanes. To explore the behaviour of a mixture of alkanes on a surface, we have carried out an MD study of the adsorption of a 50 : 50 mixture of *n*-hexadecane and HMN at 350 K and compared the structure of the mixed film with respect to films of pure components.⁷⁶

The starting configuration for the mixture was a well equilibrated liquid containing 72 molecules each of the two isomers. The bulk liquid simulation was equilibrated for 1.5 ns. We did not find any evidence of phase separation in the bulk. Later, this liquid

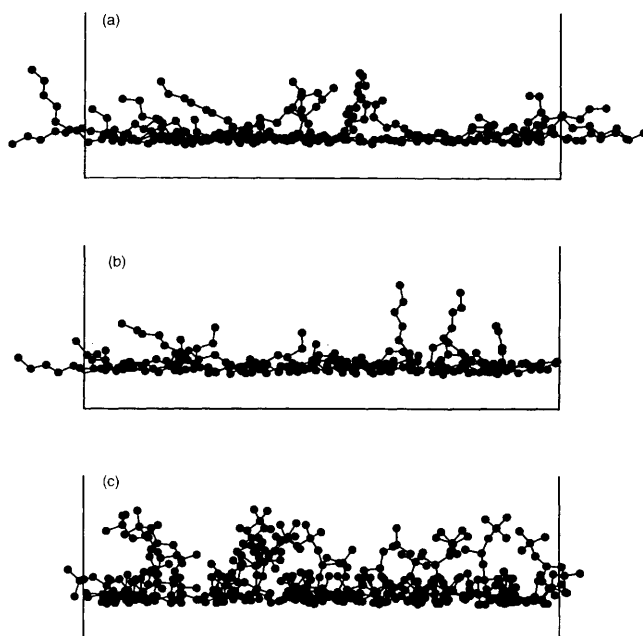


Fig. 12 Interdigitation of molecules belonging to the first layer of 40 Å thick films of isomers of hexadecane at 350 K. Side view of a snapshot of molecules belonging to the first layer. The horizontal line indicates the Au(111) of side length 43.275 Å. (a) *n*-HD; (b) PUD; (c) HMN.

was placed on an Au(111) surface. On the surface, the film was equilibrated for 900 ps, and averages were calculated over an additional run of 200 ps. In Fig. 13, we show the bead density profiles for both isomers. We observe that *n*-hexadecane molecules have been preferentially adsorbed on the surface over the HMN molecules. As we have seen earlier, the linear isomers pack better on the surface and they are thus able to increase

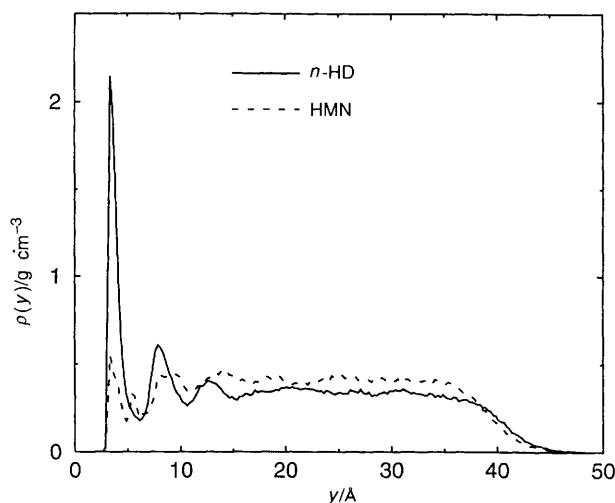


Fig. 13 Density profiles of beads of *n*-HD and HMN in an equimolar mixture film on Au(111) at 350 K. The linear isomer selectively wets the surface while the branched molecules segregate to the bulk.

their surface energy. In this case, it looks as if this increase is not offset by the decrease in the entropy of the system due to the partial segregation. This finding corroborates the earlier results of Xia and Landman on films of a mixture of $C_{16}H_{34}$ and C_6H_{14} .⁷⁵

4.2 NEMD studies

The previous section illustrates how molecular fluids exhibit density oscillations in the presence of a surface. It is natural to ask to what extent these oscillations affect shear flow. Specifically, we would like to determine the average distance from a surface at which the fluid behaves hydrodynamically. The obvious way to characterize this 'boundary layer' is by using simulations.

The problem of evaluating the hydrodynamic parameters of a solid/liquid interface was addressed by Bocquet and Barrat (BB).^{77,78} They proposed that these parameters are similar to transport coefficients, which could be calculated from equilibrium correlation functions. BB considered a geometry where a fluid slab is confined between two walls in the xz plane. A planar Couette shear field can be set up in the fluid by moving the walls with equal speeds in opposite directions along the x -axis. Navier–Stokes equations in conjunction with appropriate boundary conditions can be solved in this geometry to obtain the velocity profile of the fluid.⁷⁹ The general boundary condition could involve a slip of the fluid against the wall. BB proposed the following general boundary condition for a solid/liquid interface,^{77,78,80}

$$\left. \frac{\partial v_x}{\partial y} \right|_{y_{\text{wall}}} = \frac{1}{\delta} v_x(y_{\text{wall}}) \quad (10)$$

where, v_x is the laboratory velocity of the fluid and y_{wall} and δ are the hydrodynamic parameters that characterize the interface. y_{wall} can be understood as an average distance away from the physical solid where hydrodynamics applies. The slipping length δ is the extent to which the fluid moves along with the wall.⁸⁰ Eqn. (10) when multiplied by the viscosity of the fluid is a condition that relates the stress at a position y_{wall} to the velocity of the fluid at y_{wall} . For ideally smooth surfaces, δ is infinite. For highly corrugated surfaces, it is, in principle, zero; however, if the surface potential contains a sufficiently strong attractive term, effects such as epitaxial locking and 'stick-slip' phenomenon arise.^{81,82} In such cases, this method cannot be applied.

One can evaluate the boundary parameters, δ and y_{wall} , by moving the walls^{81,82} and fitting the steady-state velocity profile to the Navier–Stokes equation along with the boundary condition [eqn. (10)]. However, the boundary parameters are not unique for a planar Couette geometry and hence flows such as Poiseuille should be considered. This point is discussed in detail in ref. 78. Another shortcoming of performing simulations where the velocity field is produced through moving boundaries is that temperature control is not rigorously defined.^{81,83} Namely, it is usual to thermostat only a component (or sometimes two) that is orthogonal to the direction of the flow. This usually leads to heating of the order of 10–15% above the set temperature. Also, when simulating a realistic surface, such as graphite, the corrugations are sufficiently small as to cause problems with equilibration of the velocity profile. This is a major issue of concern because one needs a steady-state profile in order to fit for the boundary parameters.

Conversely, one could construct equilibrium expressions through the Mori–Zwanzig formalism for these parameters.⁷⁸ The Mori–Zwanzig theory can be used to derive a set of GK equations, which relate the boundary parameters to equilibrium correlation functions. The salient feature of this formalism is that, in principle, one can solve for the boundary parameters uniquely. This is because the GK is a linear theory that relates the fluctuations about equilibrium to a transport coefficient (or in this case a boundary parameter). Because the fluctuations about equilibrium are indistinguishable from a relaxation close to equilibrium from any mechanical perturbation, in principle, these

fluctuations contain information about all flows. Thus, uniqueness of the boundary parameters is not an issue. BB first derived the GK formulae for the boundary parameters y_{wall} and $\lambda_{\text{wall}} \equiv \eta/\delta$, where λ_{wall} is the friction coefficient and η is the shear viscosity.⁷⁸ The GK formulae are,

$$y_{\text{wall}} = \int_0^\infty dt \langle F_x(t) \sigma_{xy}(0) \rangle_c / \int_0^\infty dt \langle F_x(t) F_x(0) \rangle_c$$

$$\lambda_{\text{wall}} = \frac{1}{S k_B T} \int_0^\infty dt \langle F_x(t) F_x(0) \rangle_c$$

Here, F_x is the force on the fluid imparted by the wall, S is the surface area, k_B is Boltzmann's constant, the subscript c denotes an average in the canonical ensemble and σ_{xy} is the xy component of the stress tensor for the fluid including the contribution from the wall.⁸⁴

It is also possible to use a set of equations of motion that, within linear response theory, would yield the equilibrium parameters. This approach is similar to using an external field to obtain the shear viscosity of bulk fluids, as discussed in Section 3.2. The perturbation to the canonical Hamiltonian representing the coupling between the external field and the fluid is given by,⁷⁸

$$H[\gamma, y_p] = \gamma \sum_{i=1}^N (y_i - y_0) p_{i,x} \quad (11)$$

This perturbation generates a stable linear velocity profile in the fluid given, to first order in γ , by $v_x(y) = \gamma(y - y_0)$.³⁷ Thus, y_0 is the position in the fluid where the field (and hence the velocity) is zero. One can now ask the question: What is the response of a confined fluid to this perturbation? If the response variable is the average (steady state) force from the walls on the moving fluid ($\langle F_x \rangle_{\text{NE}}$), the linear response theory with eqn. (11) as the perturbation yields,⁷⁸

$$\langle F_x \rangle_{\text{NE}} = -S \lambda_{\text{wall}} v_x(y_{\text{wall}}) \quad (12)$$

where the definitions for y_{wall} and λ_{wall} are preserved. Thus BB found that the DOLLS perturbation returns the GK relations derived in eqn. (11), within linear response, when the solid walls are stationary. Eqn. (12) is seen to be similar to the general boundary condition eqn. (10) in that it relates the stress on the fluid to the velocity at the hydrodynamic boundary.

In the practical implementation of eqn. (12), we have used the SLLOD equations of motion, eqn. (7), without reference to a boundary condition (*e.g.* Lees–Edwards). One important distinction between the SLLOD equations and the DOLLS perturbation eqn. (11) is that SLLOD produces stable planar-Couette flow for all values of γ .^{37,85} Since it is imperative that a linear velocity profile be maintained, it is useful to show that the SLLOD equations, implemented without reference to boundary conditions, will achieve this. It has been shown that if one supplements eqn. (7) with an additional redundant variable $\dot{I} = \gamma p_x r_y p_z / Q$, one can derive a conserved quantity for SLLOD dynamics in the absence of boundary conditions.⁸⁵ Besides being a useful check on the algorithm, the conserved quantity is the starting point for a phase-space analysis. We have shown that SLLOD dynamics, without boundary conditions, produces a stable linear profile when every degree of freedom is thermostatted (the so-called ‘massive’ thermostating scheme).^{85,86} With proof that eqn. (7) generates a field-driven profile, it is possible to determine the response of F_x to the independent variables γ and y_0 .

When eqn. (12) is used to derive the boundary parameters, care should be taken to ensure that the simulations are performed within the limits of linear response theory. This is a necessary condition to obtain parameters that correspond to those obtained

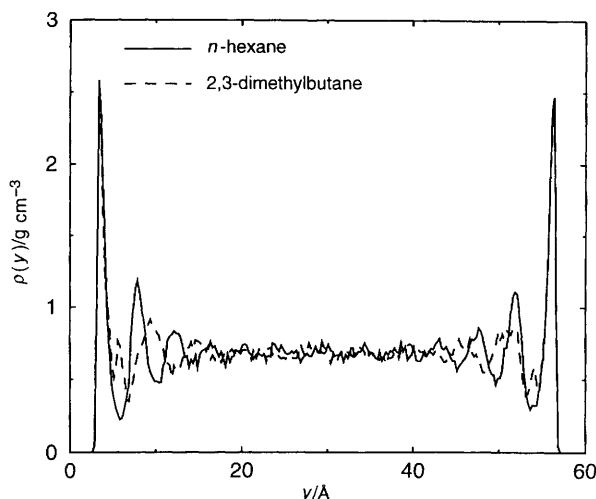


Fig. 14 Bead density profiles of *n*-hexane and 2,3-dimethylbutane confined between graphite surfaces separated by 60 Å at zero shear

from equilibrium simulation using eqn. (11). It should be noted that the perturbation for shearing a confined fluid depends not only on γ but also on y_0 . The linear response expression eqn. (12) is valid only for proper choices of the set $\{\gamma, y_0\}$.⁸⁴ This is in contrast to simulations performed in bulk to calculate the shear viscosity, where there is no dependence on y_0 .

In Section 4.1, we had seen that branching affects significantly the layering of molecules on a surface. In a similar fashion, we calculate here the interfacial parameters for two isomers of C_6H_{14} under identical conditions: *n*-hexane and 2,3-dimethylbutane.

An anisotropic graphite model has been employed for the alkane–surface interactions.⁸⁷ The simulation system consisted of 240 alkane molecules at a density of 0.65 g cm^{-3} between graphite surfaces in the x – z plane. The surfaces were separated in the y -direction by 60 Å. Such a large separation is essential for eqn. (12) to be applied. The simulations were performed at 325 K. The confined fluids were equilibrated for 600 ps. The zero-shear density profiles of the two fluids are shown in Fig. 14. The density profiles show nearly 3 to 4 layers on either surface in agreement with the equilibrium results presented in Section 4.1.

To generate a steady linear velocity profile in the fluid, it was necessary to employ massive thermostating.⁸⁶ The linear response regime for F_x to $v_x(y_{\text{wall}})$ was determined by performing several runs at different values of y_0 . Hydrodynamic boundary parameters were extracted from such runs using eqn. (12). The value of y_{wall} reflects the behaviour of the molecules in the first layer as seen from their density profiles. For 2,3-dimethylbutane, we found $y_{\text{wall}} = 5.1 \pm 0.1 \text{ Å}$ and for *n*-hexane, $y_{\text{wall}} = 4.1 \pm 0.1 \text{ Å}$. The values of y_{wall} , interpreted as an ‘immobile layer thickness,’ are in qualitative agreement with the experiments of Georges *et al.*⁸⁸ The friction coefficient, λ_{wall} , a more difficult quantity to calculate, is seemingly independent of the molecular architecture. Calculation of a slipping length from this quantity yields a value of *ca.* 80 Å, which is consistent with the estimates of Padilla and Toxvaerd.⁸³

5 Conclusions

We have presented a self-consistent set of transferable potential parameters for alkane molecules fitted to bulk properties. These potentials have allowed us to proceed with

confidence in studying transport and structural properties, in bulk and under confinement, of more complicated alkane architectures. The ramifications of molecular architecture on the properties of thin liquid alkane films adsorbed on an Au(111) surface were explored. We have used the theory of Bocquet and Barrat for fluids in confined geometries, in conjunction with our potentials, to provide a microscopic description of hydrodynamic boundary parameters that characterize the solid/liquid interface of liquid alkanes on graphite. Where comparisons between our simulations and experiment are possible, excellent agreement is achieved. The simulation methodologies have been sufficiently successful that the investigation of more complex systems, including mixtures, can be envisioned.

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References

- 1 J. N. Israelachvili, *Intermolecular and Surface Forces*, Academic, New York, 1985.
- 2 J. N. Israelachvili, P. M. McGuiggan and A. M. Homola, *Science*, 1988, **240**, 189.
- 3 H. K. Christenson, D. W. R. Gruen, R. G. Horn and J. N. Israelachvili, *J. Chem. Phys.*, 1989, **87**, 1834.
- 4 S. K. Kumar, M. Vacatello and D. Y. Yoon, *J. Chem. Phys.*, 1988, **89**, 5209; M. Vacatello, D. Y. Yoon and B. C. Laskowski, *J. Chem. Phys.*, 1990, **93**, 779.
- 5 M. L. Gee, P. M. McGuiggan, J. N. Israelachvili and A. M. Homola, *J. Chem. Phys.*, 1990, **93**, 1895.
- 6 P. A. Thompson and M. O. Robbins, *Science*, 1990, **250**, 792.
- 7 S. Granick, *Science*, 1991, **253**, 1374.
- 8 S. Granick, in *Fundamentals of Friction: Macroscopic and Microscopic Processes*, NATO ASI Conf. Proc. Ser. E, Vol. 220, ed. I. L. Singer and H. M. Pollock, Kluwer Academic, New York, 1992.
- 9 M. Schoen, C. L. Rhykerd Jr., D. J. Diestler and J. H. Cushman, *Science*, 1989, **245**, 1223; M. Schoen, D. J. Diestler and J. H. Cushman, *J. Chem. Phys.*, 1987, **87**, 5464; M. Schoen, D. J. Diestler and J. H. Cushman, *J. Chem. Phys.*, 1994, **101**, 6865.
- 10 T. K. Xia, J. Quyang, M. W. Ribarsky and U. Landman, *Phys. Rev. Lett.*, 1992, **69**, 1967.
- 11 J. G. Harris, *J. Phys. Chem.*, 1992, **96**, 5077.
- 12 M. W. Ribarsky and U. Landman, *J. Chem. Phys.*, 1992, **97**, 1937.
- 13 T. K. Xia and U. Landman, *Science*, 1993, **261**, 1310; *Phys. Rev. B*, 1993, **48**, 11313.
- 14 H. Yoshizawa, Y.-L. Chen and J. Israelachvili, *J. Phys. Chem.*, 1993, **97**, 4128; H. Yoshizawa and J. Israelachvili, *J. Phys. Chem.*, 1993, **97**, 11300.
- 15 Y. Wang, Y. K. Hill and J. G. Harris, *J. Chem. Phys.*, 1994, **100**, 3276.
- 16 B. Bhushan, J. N. Israelachvili and U. Landman, *Nature (London)*, 1995, **374**, 607.
- 17 S. Balasubramanian, M. L. Klein and J. I. Siepmann, *J. Chem. Phys.*, 1995, **103**, 3184.
- 18 R. E. Reiher, Ph.D. Thesis, Harvard University, Cambridge, MA, 1985.
- 19 P. van der Ploeg and H. J. C. Berendsen, *J. Chem. Phys.*, 1982, **76**, 3271.
- 20 W. L. Jorgenson, J. D. Madura and C. J. Swenson, *J. Am. Chem. Soc.*, 1984, **106**, 6638.
- 21 J. I. Siepmann, S. Karaborni and B. Smit, *Nature (London)*, 1993, **365**, 330.
- 22 B. Smit, S. Karaborni and J. I. Siepmann, *J. Chem. Phys.*, 1995, **102**, 2126.
- 23 A. Z. Panagiotopoulos, *Mol. Phys.*, 1987, **61**, 813.
- 24 N. Quirke, M. Stapleton and D. J. Tildesley, *Mol. Phys.*, 1988, **63**, 527.
- 25 B. Smit, P. de Smedt and D. Frenkel, *Mol. Phys.*, 1989, **68**, 931.
- 26 A. Z. Panagiotopoulos, *Mol. Sim.*, 1992, **9**, 1.
- 27 J. I. Siepmann, C. J. Mundy and M. L. Klein, 1997, *Mol. Phys.*, in the press.
- 28 B. D. Smith and R. Srivastava, *Thermodynamics of Pure Compounds: Hydrocarbons and Ketones*, Elsevier, Amsterdam, 1986.
- 29 J. I. Siepmann, *Mol. Phys.*, 1990, **70**, 1145.
- 30 J. I. Siepmann, J. I. and D. Frenkel, *Mol. Phys.*, 1992, **75**, 59; J. I. Siepmann, in *Computer Simulation of Biomolecular Systems: Theoretical and Experimental Applications*, ed. W. F. van Gunsteren, P. K. Weiner and A. J. Wilkinson, Escom Science, 1993.
- 31 D. Frenkel, G. C. A. M. Mooij and B. Smit, *J. Phys.: Condens. Matter*, 1992, **4**, 3053.
- 32 J. J. de Pablo, M. Laso and U. W. Suter, *J. Chem. Phys.*, 1992, **97**, 2817.
- 33 G. J. Martyna, D. J. Tobias and M. L. Klein, *J. Chem. Phys.*, 1994, **101**, 4177.
- 34 D. M. Small, *The Physical Chemistry of Lipids*, Plenum, New York, 1988.
- 35 M. Mondello and G. S. Grest, *J. Chem. Phys.*, 1995, **103**, 7156.

- 36 *Properties of Hydrocarbons of High Molecular Weight*, American Petroleum Institute, Research Project 42, Washington DC, 1966.
- 37 D. J. Evans and G. P. Morriss, in *Statistical Mechanics of Nonequilibrium Liquids*, Academic, London, 1986.
- 38 M. P. Allen and D. J. Tildesley, *Computer Simulation of Liquids*, Oxford University Press, Oxford, 1987.
- 39 S. Nosé, *J. Chem. Phys.*, 1984, **81**, 511.
- 40 W. G. Hoover, *Phys. Rev. A*, 1985, **31**, 1695.
- 41 G. J. Martyna, M. L. Klein and M. E. Tuckerman, *J. Chem. Phys.*, 1992, **97**, 2635.
- 42 M. E. Tuckerman, B. J. Berne and G. J. Martyna, *J. Chem. Phys.*, 1992, **97**, 1990.
- 43 C. J. Mundy, J. I. Siepmann and M. L. Klein, *J. Chem. Phys.*, 1995, **102**, 3376.
- 44 S. T. Cui, P. T. Cummings and H. D. Cochran, *J. Chem. Phys.*, 1996, **104**, 255.
- 45 The experimental value for the viscosity of *n*-decane in this figure and Table 4 has been adjusted from the value in ref. 43 and it is now the same as in ref. 44. We are grateful to Peter Cummings for discovering a missing factor of two in our long-range corrections to the pressure.
- 46 C. J. Mundy, J. I. Siepmann and M. L. Klein, to be published.
- 47 K. Stephan and K. Lucas, *Viscosity of Dense Fluids*, Plenum, New York, 1979.
- 48 C. Barus, *Am. J. Sci.*, 1893, **45**, 87.
- 49 D. Cooper and A. J. Moore, *Wear*, 1994, **175**, 93.
- 50 P. J. Davis and D. J. Evans, *J. Chem. Phys.*, 1994, **100**, 54.
- 51 C. J. Mundy, J. I. Siepmann and M. L. Klein, *J. Chem. Phys.*, 1995, **103**, 10192.
- 52 Z. Xu, J. J. de Pablo and S. Kim, *J. Chem. Phys.*, 1995, **102**, 5836.
- 53 J. N. Israelachvili and G. E. Adams, *J. Chem. Soc., Faraday Trans.*, 1978, **174**, 975.
- 54 S. A. Somers and H. T. Davis, *J. Chem. Phys.*, 1992, **96**, 5389.
- 55 I. A. Bitsanis and C. Pan, *J. Chem. Phys.*, 1993, **99**, 5520.
- 56 R. G. Winkler, A. Gerstmaier, P. Reineker, T. Matsuda and D. Y. Yoon, *Int. J. Quantum Chem.*, 1994, **52**, 437.
- 57 P. Padilla and S. Toxvaerd, *J. Chem. Phys.*, 1994, **101**, 1490.
- 58 P. A. Thompson, M. O. Robbins and G. S. Grest, *Israel J. Chem.*, 1995, **35**, 93.
- 59 S. Granick, A. L. Demirel, L. L. Cai and J. Peanasky, *Israel J. Chem.*, 1995, **35**, 75.
- 60 S. Balasubramanian, M. L. Klein and J. I. Siepmann, *J. Phys. Chem.*, 1996, **100**, 11660.
- 61 H.-W. Hu, G. A. Carson and S. Granick, *Phys. Rev. Lett.*, 1991, **66**, 2758.
- 62 G. Reiter, A. L. Demirel, J. Peanasky, L. L. Cai and S. Granick, *J. Chem. Phys.*, 1994, **101**, 2606.
- 63 M. Schoen, S. Hess and D. J. Diestler, *Phys. Rev. E*, 1995, **52**, 2587.
- 64 A. J. Groszek, *Proc. R. Soc. London, Series A*, 1970, **314**, 473.
- 65 L. Battezzati, C. Pisani and F. Ricca, *J. Chem. Soc., Faraday Trans.*, 1975, **71**, 1629.
- 66 G. C. McGonigal, R. H. Bernhardt and D. J. Thomson, *Appl. Phys. Lett.*, 1990, **57**, 28; G. C. McGonigal, R. H. Bernhardt, Y. H. Yeo and D. J. Thomson, *J. Vac. Sci. Technol. B*, 1991, **9**, 1107.
- 67 J. P. Rabe and S. Buchholz, *Science*, 1991, **253**, 424; L. Askadskaya and J. P. Rabe, *Phys. Rev. Lett.*, 1992, **69**, 1395.
- 68 G. Watel, F. Thibaudau and J. Cousty, *Surf. Sci. Lett.*, 1993, **281**, L297.
- 69 S. Balasubramanian, J. I. Siepmann and M. L. Klein, unpublished results.
- 70 W. A. Steele, *Surf. Sci.*, 1973, **36**, 317; W. A. Steele, A. V. Vernov and D. J. Tildesley, *Carbon*, 1987, **25**, 7.
- 71 S. Leggetter and D. J. Tildesley, *Mol. Phys.*, 1989, **68**, 519.
- 72 K. W. Herwig, B. Matthies and H. Taub, *Phys. Rev. Lett.*, 1995, **75**, 3154.
- 73 Y. Wang, K. Hill and J. G. Harris, *Langmuir*, 1993, **9**, 1983.
- 74 J. N. Israelachvili, S. J. Kott, M. Gee and T. A. Witten, *Macromolecules*, 1989, **22**, 4247; M. L. Gee and J. N. Israelachvili, *J. Chem. Soc., Faraday Trans.*, 1990, **86**, 4049.
- 75 T. K. Xia and U. Landman, *Science*, 1993, **261**, 1310.
- 76 S. Balasubramanian, C. J. Mundy and M. L. Klein, unpublished results.
- 77 L. Bocquet and J.-L. Barrat, *Phys. Rev. Lett.*, 1993, **70**, 2726.
- 78 L. Bocquet and J.-L. Barrat, *Phys. Rev. E*, 1994, **49**, 3079.
- 79 D. J. Tritton, *Physical Fluid Dynamics*, Clarendon Press, Oxford, 1988.
- 80 P. G. Wolynes, *Phys. Rev. A*, 1976, **13**, 1235.
- 81 P. A. Thompson and M. O. Robbins, *Phys. Rev. A*, 1990, **41**, 6830.
- 82 P. A. Thompson, G. S. Grest and M. O. Robbins, *Phys. Rev. Lett.*, 1992, **68**, 3448.
- 83 P. Padilla and S. Toxvaerd, *J. Chem. Phys.*, 1994, **101**, 1490.
- 84 C. J. Mundy, S. Balasubramanian and M. L. Klein, *J. Chem. Phys.*, in the press.
- 85 M. E. Tuckerman, C. J. Mundy, S. Balasubramanian and M. L. Klein, *J. Chem. Phys.*, 1996, **105**, 3211.
- 86 D. J. Tobias, G. J. Martyna and M. L. Klein, *J. Phys. Chem.*, 1993, **97**, 12959.
- 87 G. Vidali and M. W. Cole, *Phys. Rev. B*, 1984, **29**, 6736; Y. P. Joshi and D. J. Tildesley, *Mol. Phys.*, 1985, **55**, 999; A. Cheng and M. L. Klein, *Langmuir*, 1992, **8**, 2798.
- 88 J. M. Georges, S. Millot, J. L. Loubet and A. Tonck, *J. Chem. Phys.*, 1993, **98**, 7345.
- 89 P. Padilla and S. Toxvaerd, *J. Chem. Phys.*, 1995, **103**, 716.